



East Coast Drifters

Project Update
9/24/2026



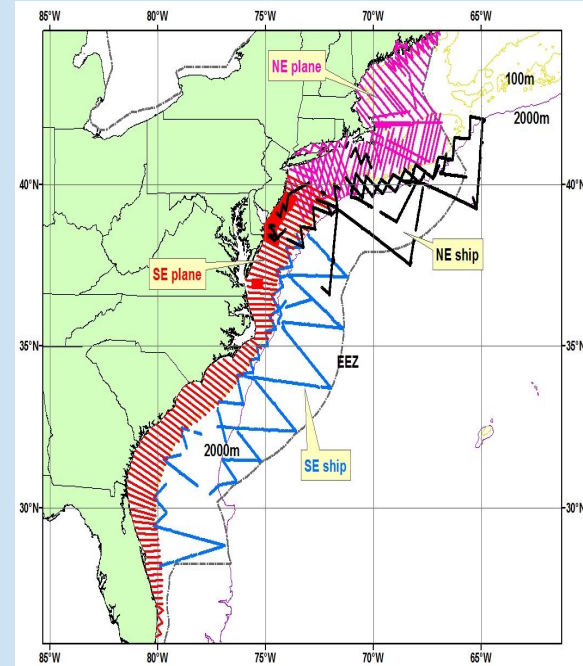
NMFS Stock Assessments in the Western North Atlantic



NOAA Technical Memorandum NMFS-NE-321

This series represents a secondary level of scientific publishing. All issues employ thorough internal scientific review; some issues employ external scientific review. Reviews are transparent collegial reviews, not anonymous peer reviews. All issues may be cited in formal scientific communications.

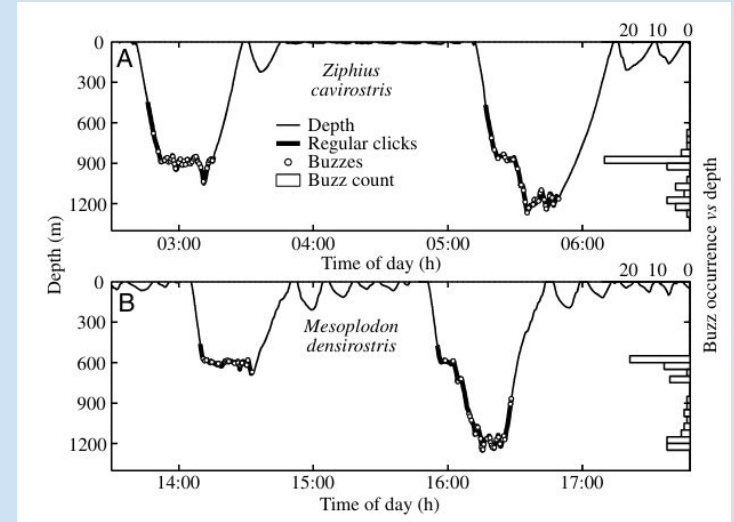
U.S. Atlantic and Gulf of Mexico Marine Mammal Stock Assessments 2023



PAM for 'cryptic species'

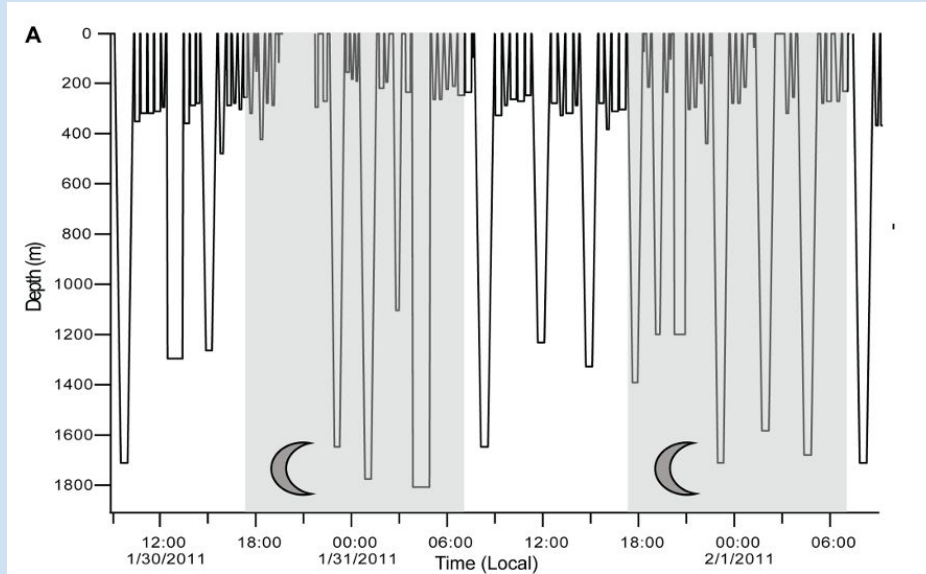


Photo credit: Lindsey Transue

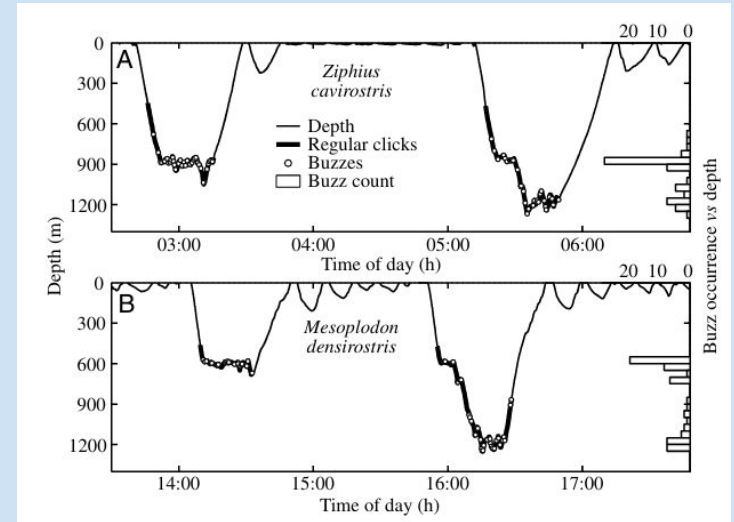


Tyack et al. 2006

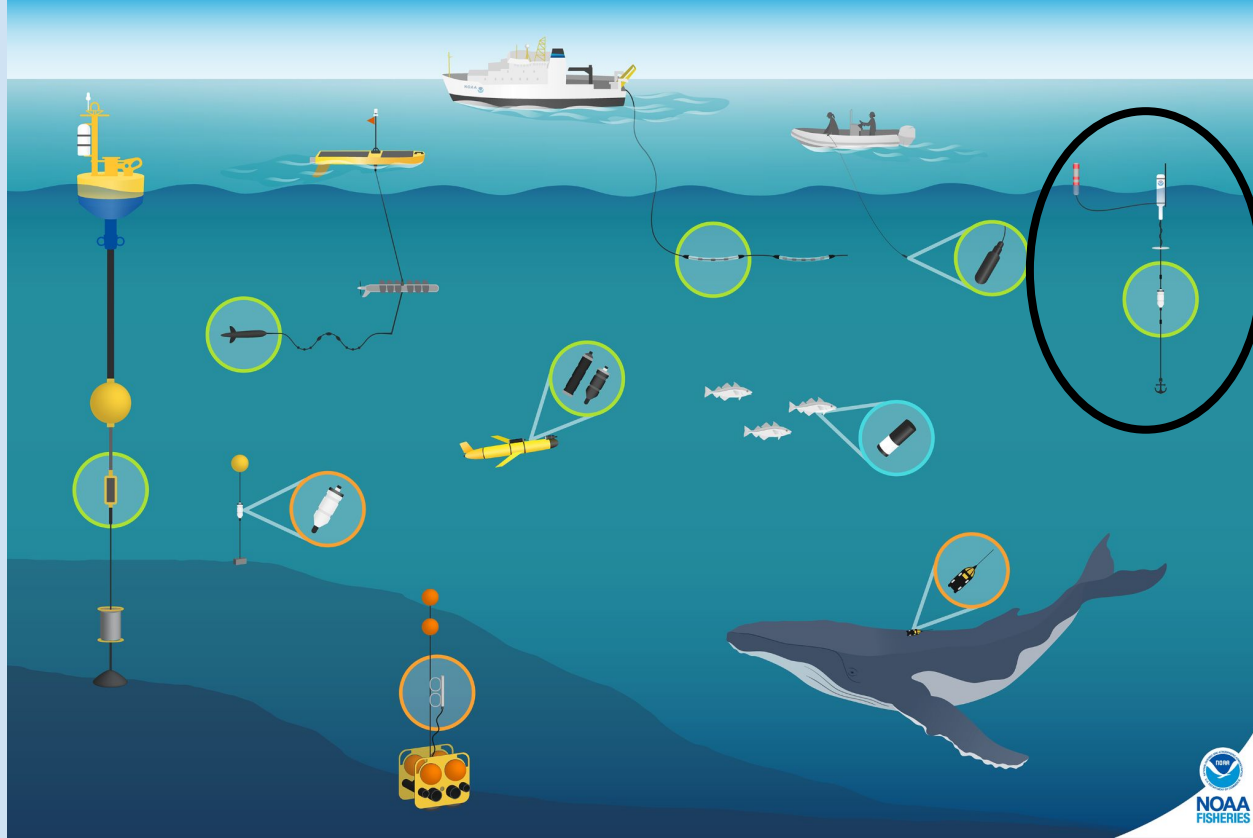
PAM for 'cryptic species'



Schorr et al. 2014

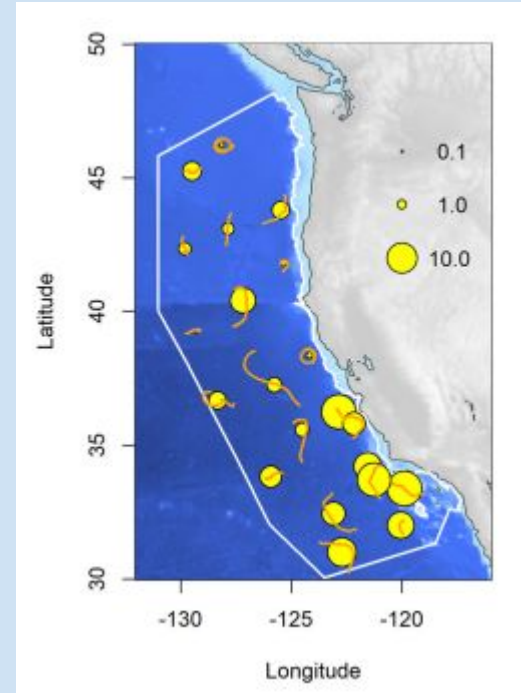
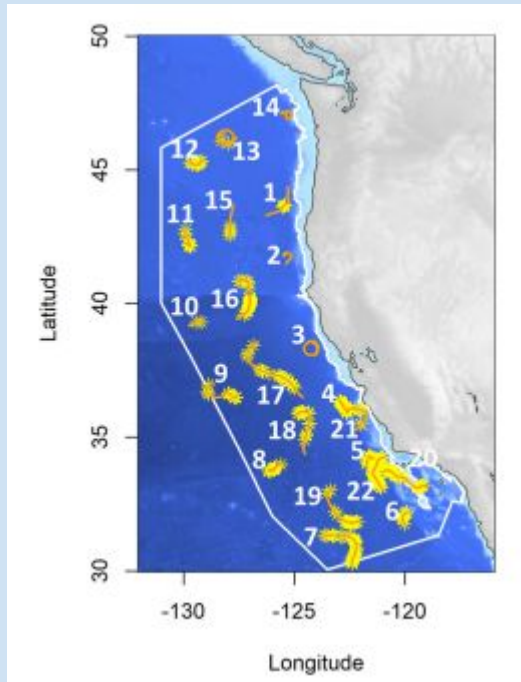


Tyack et al. 2006

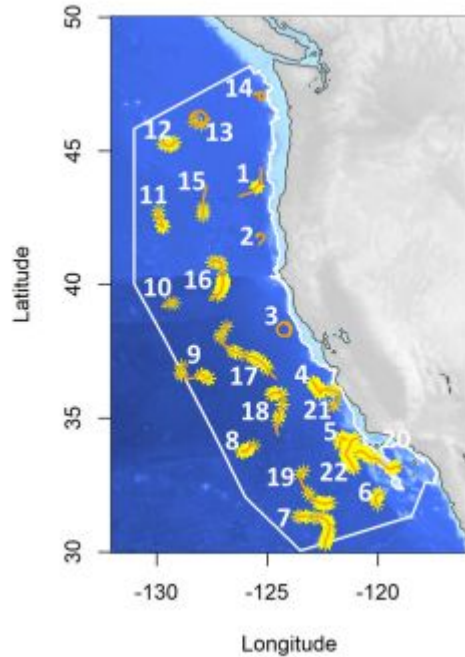


- ✓ Slow travel speed
- ✓ Good spatial coverage
- ✓ Easy to assemble and deploy
- ✓ Quiet
- ✓ Low cost

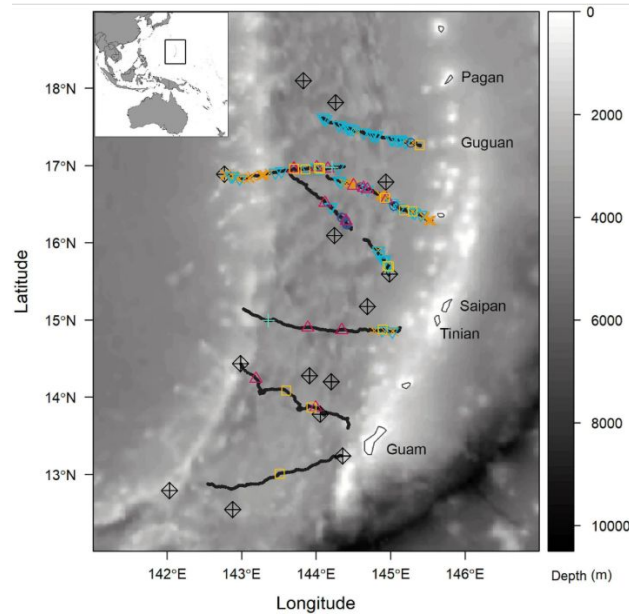
From distribution to density



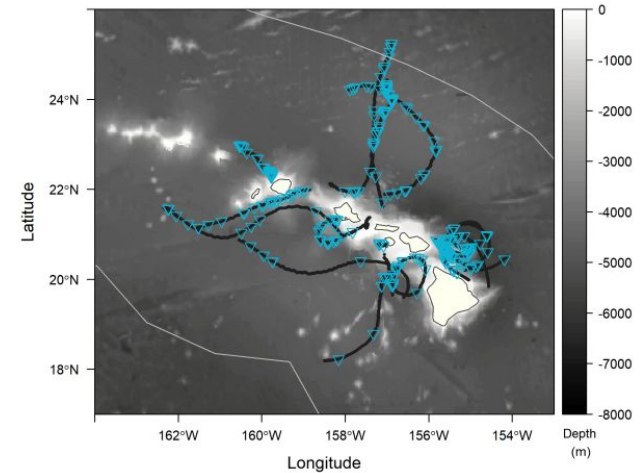
Drifting for beaked whales



Barlow et al. 2021

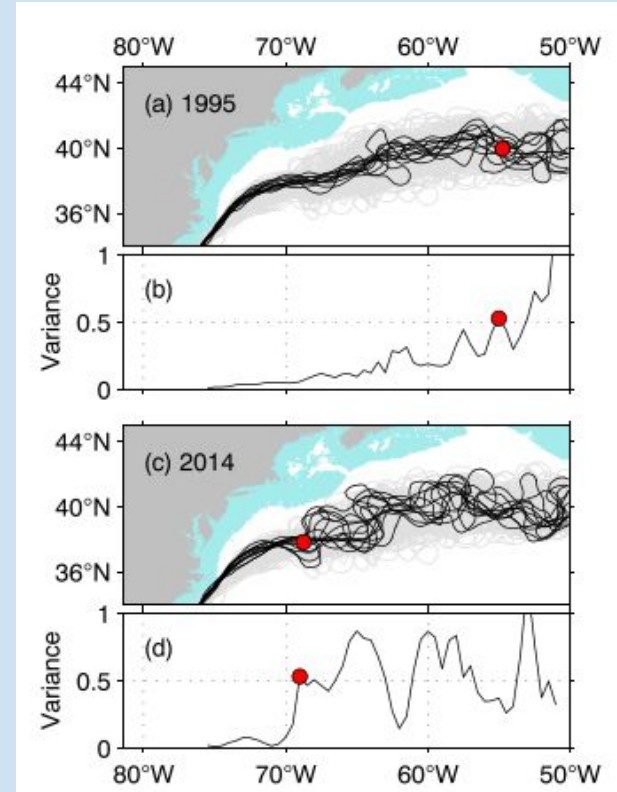
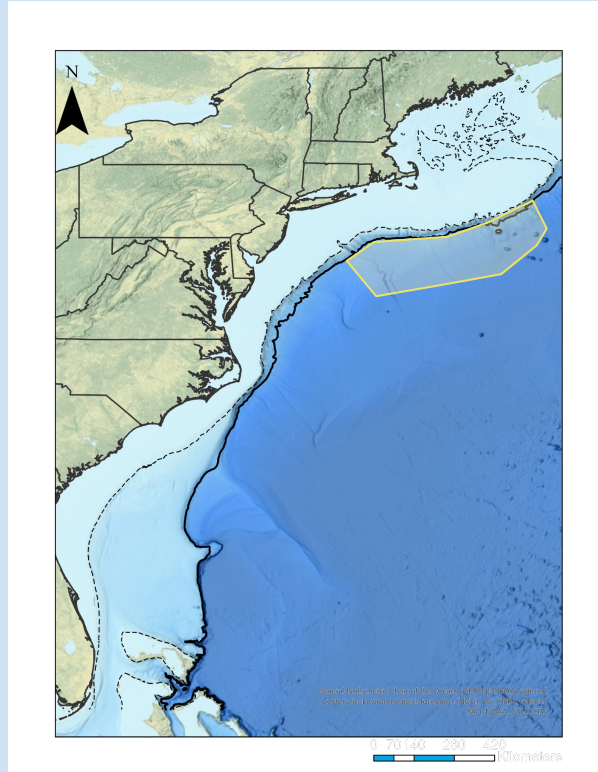


McCullough et al. 2021a

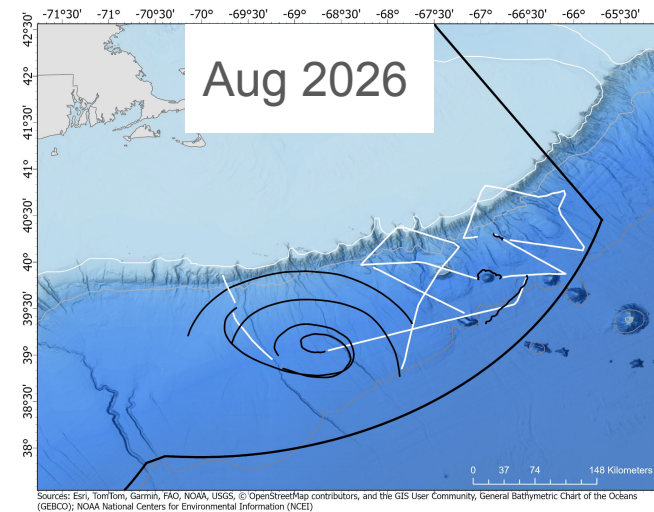
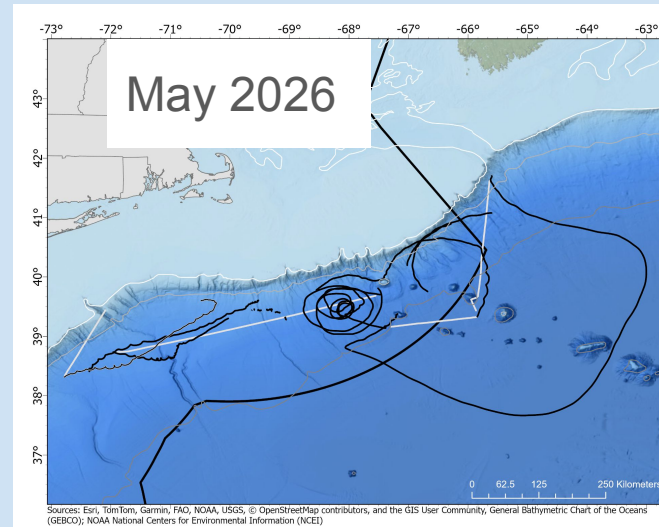
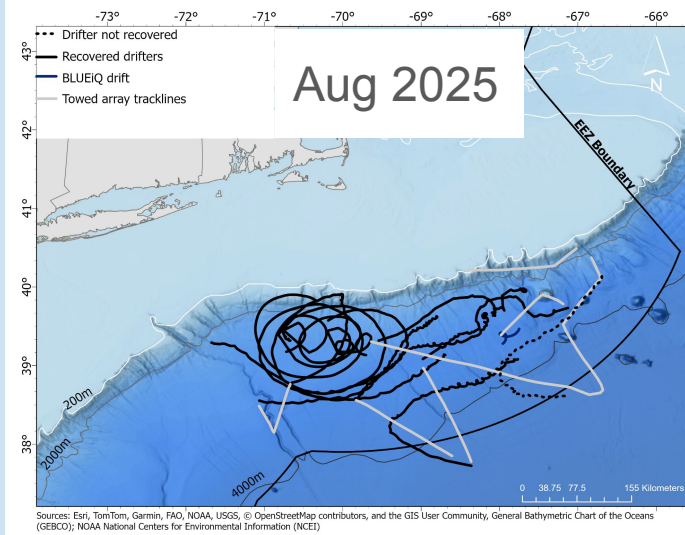


McCullough et al. 2021b

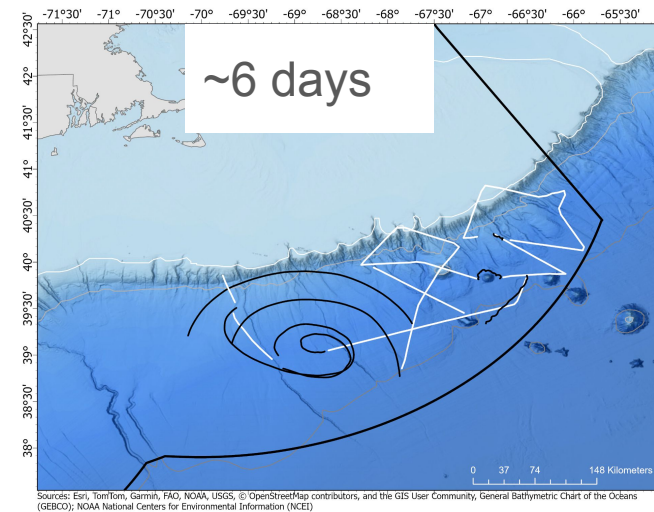
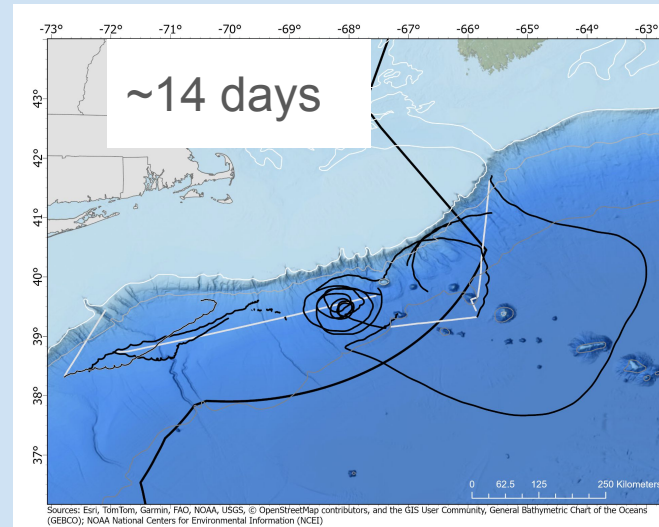
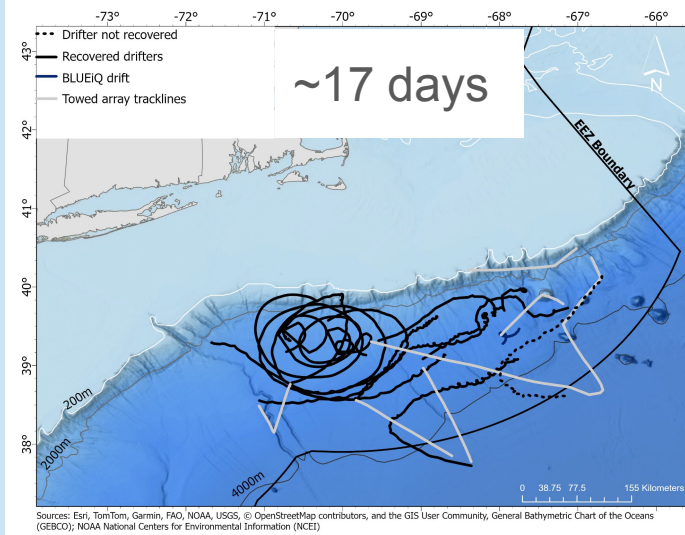
Expand to the US East Coast



PAM SI: 3 Field Seasons

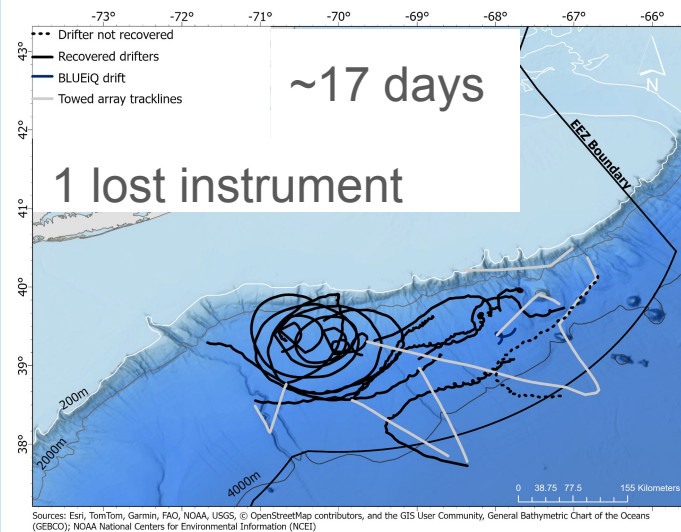


PAM SI: 3 Field Seasons

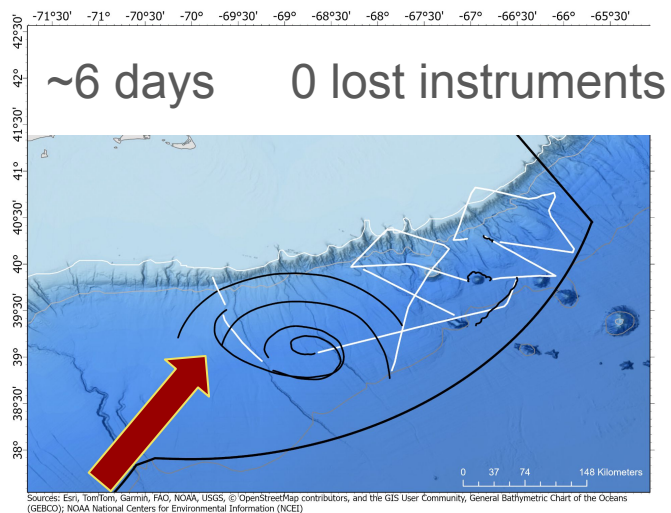
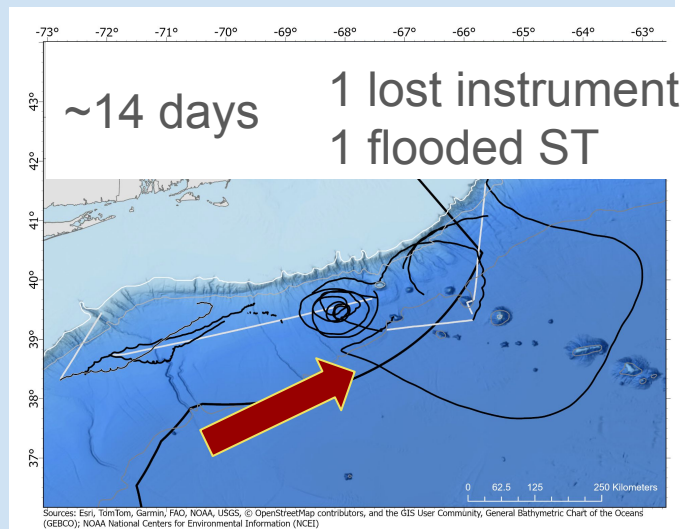


PAM SI: 3 Field Seasons

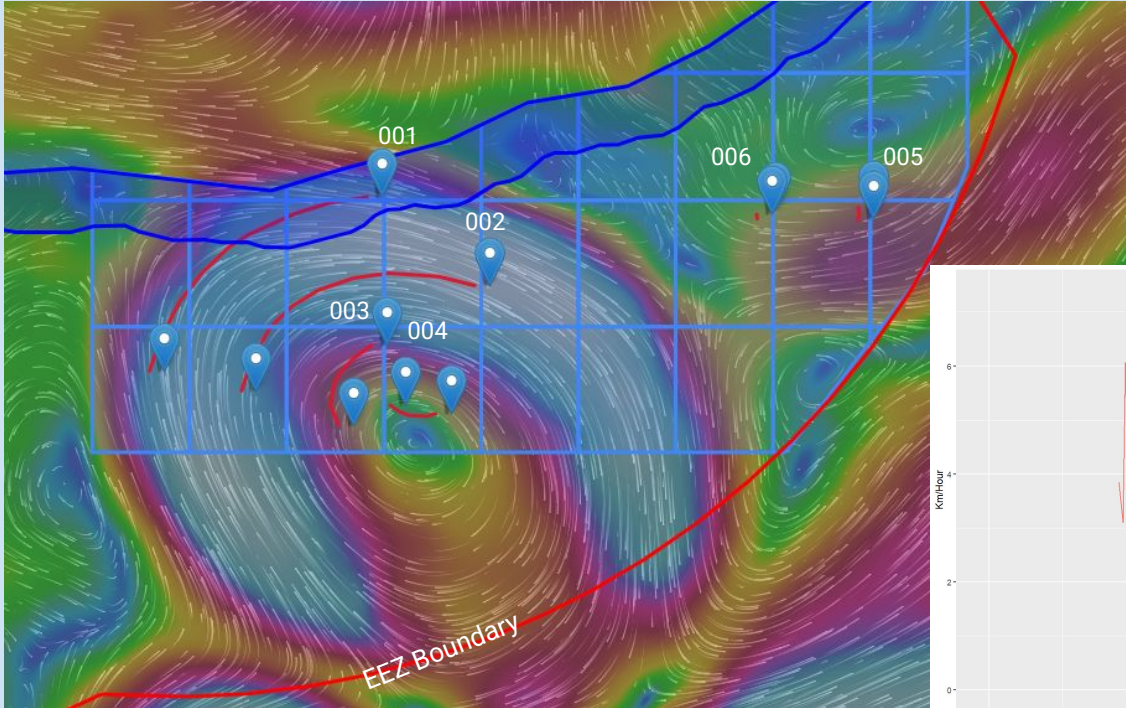
No Gulf Stream



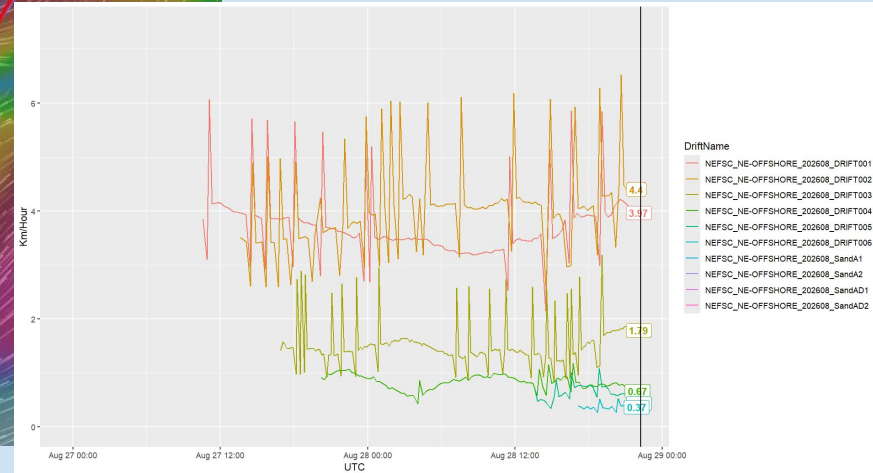
Gulf Stream



202608 field season



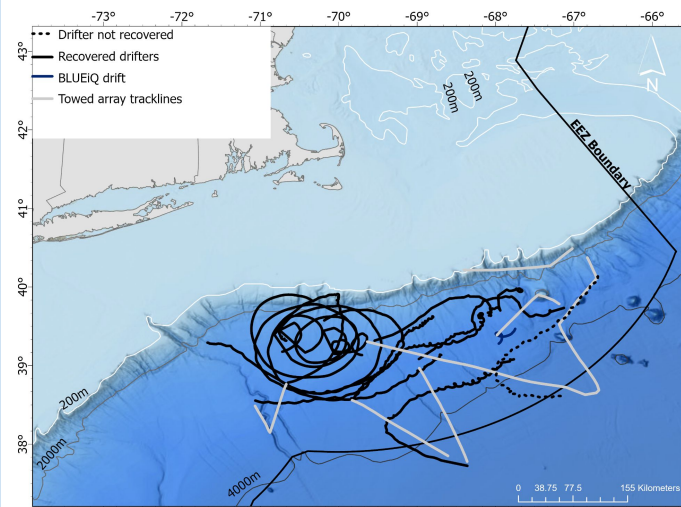
Drifts 1 and 2 going 4-4.5 km/hour



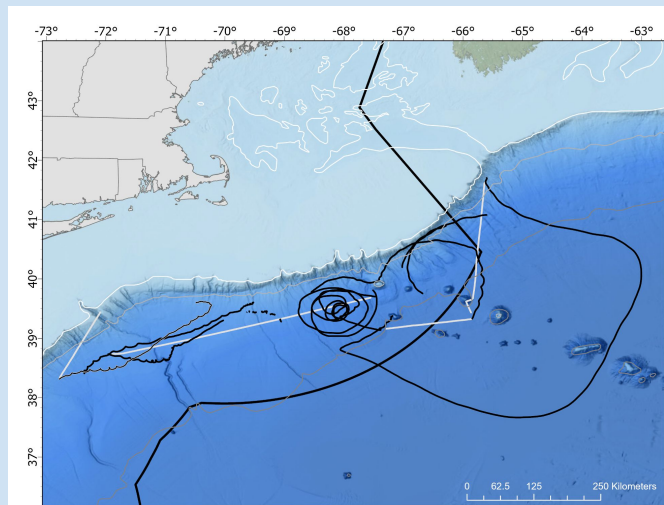
Lessons learned

★ Gulf Stream placement critical to mission

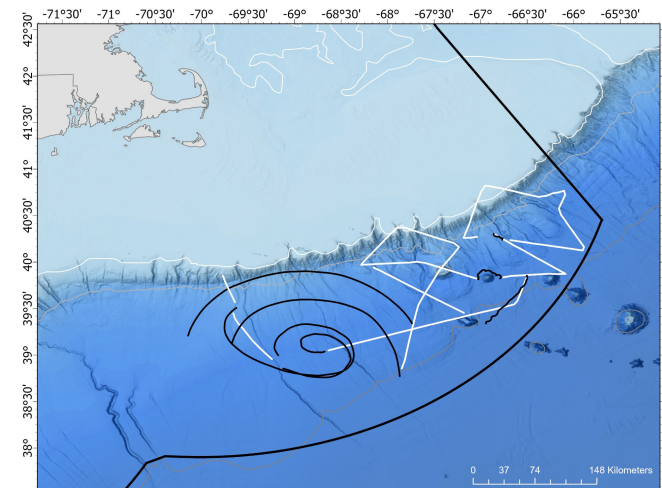
- If within the study area, plan for survey to stay out for data collection
 - Shorter study (~1 week)
- If south of study area, can deploy and leave drifters for ~2 weeks



Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community, General Bathymetric Chart of the Oceans (GEBCO); NOAA National Centers for Environmental Information (NCEI)



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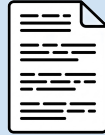
Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community, General Bathymetric Chart of the Oceans (GEBCO); NOAA National Centers for Environmental Information (NCEI)

Mooring design

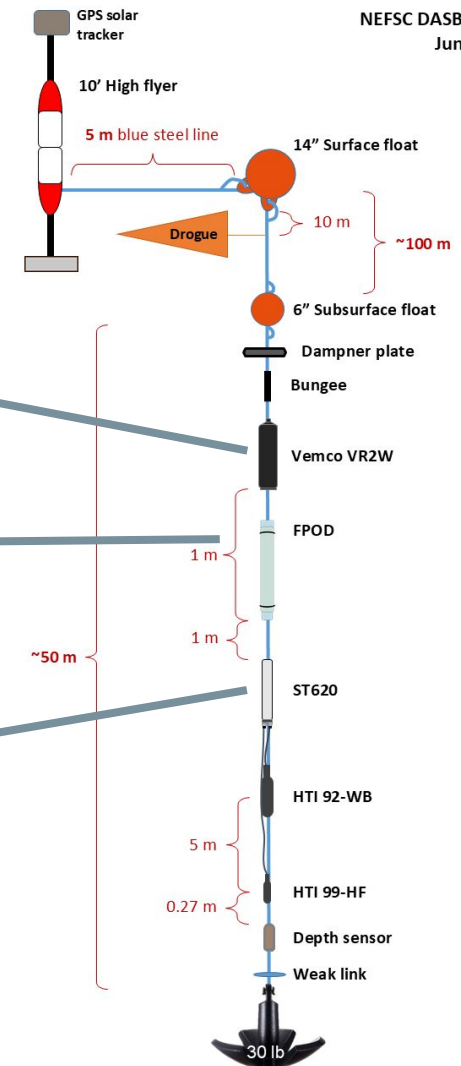
VEMCO



FPOD



SoundTrap 620

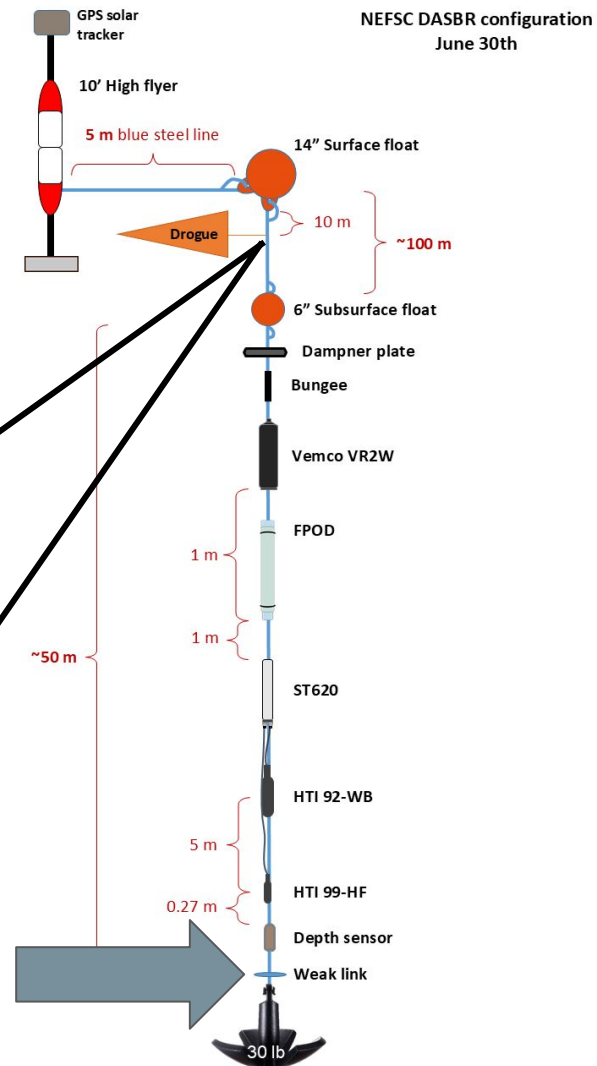


Optimizing the mooring for the western North Atlantic

Add weak rope every 60' to prevent entanglement

(Novabraid South Shore Sleeve)

Tested Aug 2026 deployment **SUCCESS**

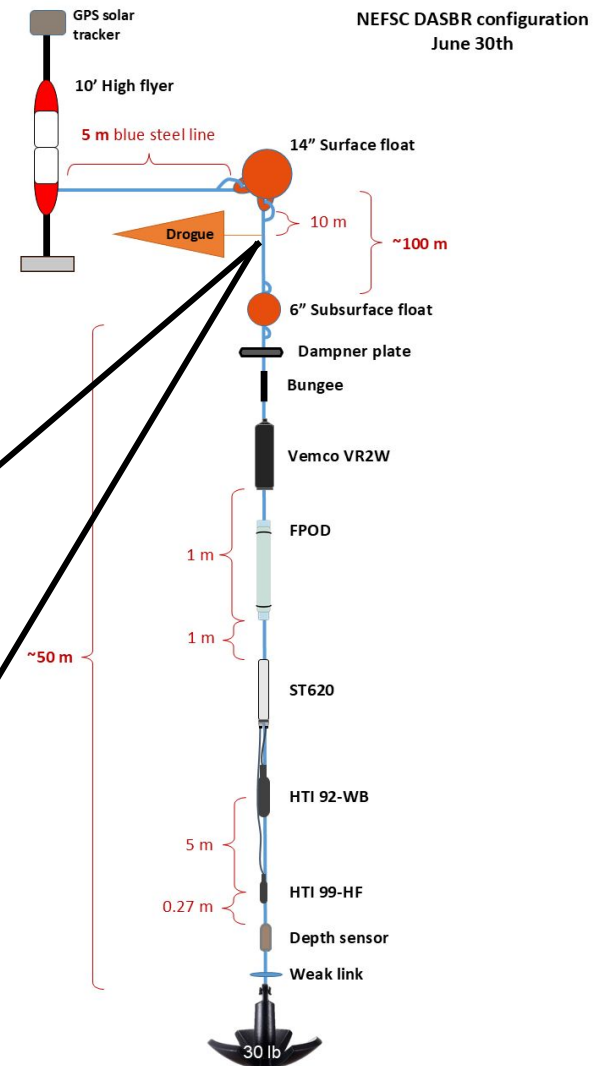


Optimizing the mooring for the western North Atlantic

Add weak link every 60' to prevent entanglement

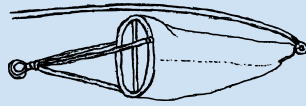
Tested Aug 2026 deployment

SUCCESS
POT HAULER

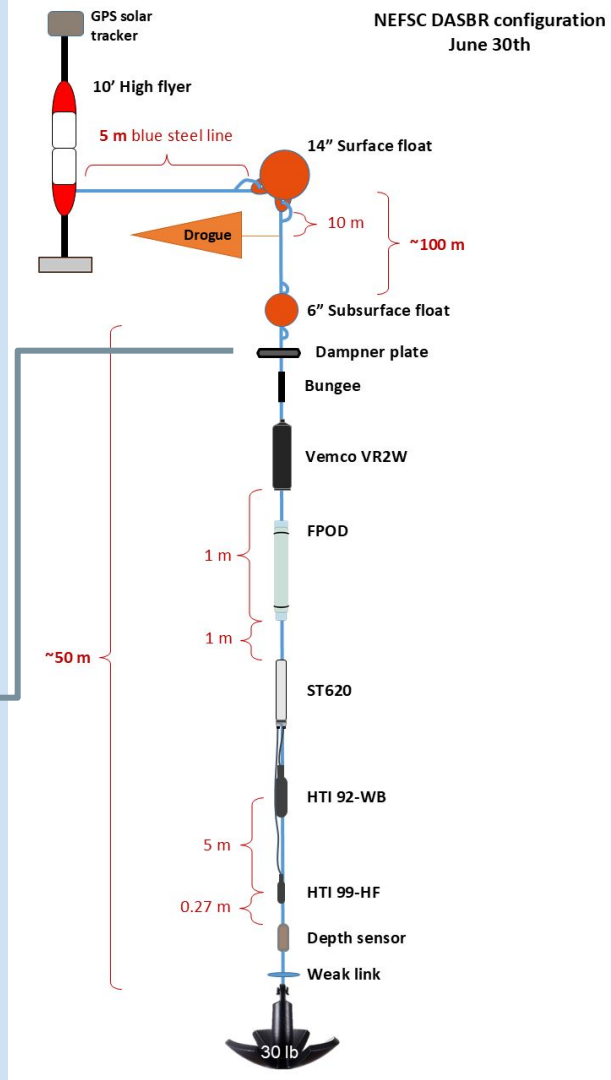
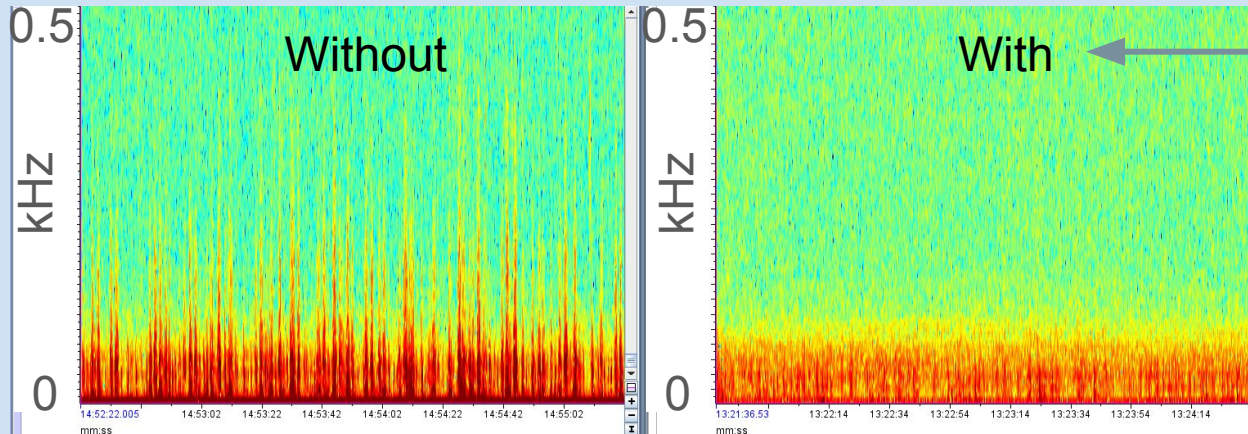


Optimizing the mooring for the western North Atlantic

Drogue (sea anchor) does not appear to slow down drift



Dampner plates are essential for baleen whale detection



Analysis goals

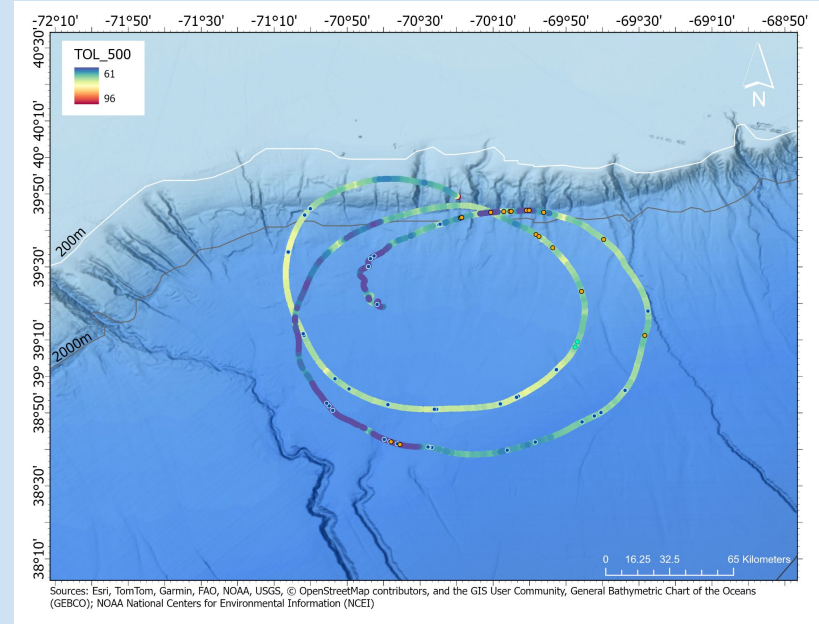
Distribution for baleen whales, beaked whales, Kogia, sperm whales, and dolphins

Density for beaked whales

Soundscapes in the deep sea

How does the soundscape relate to
beaked whales?

How does the soundscape compare to
on shelf?



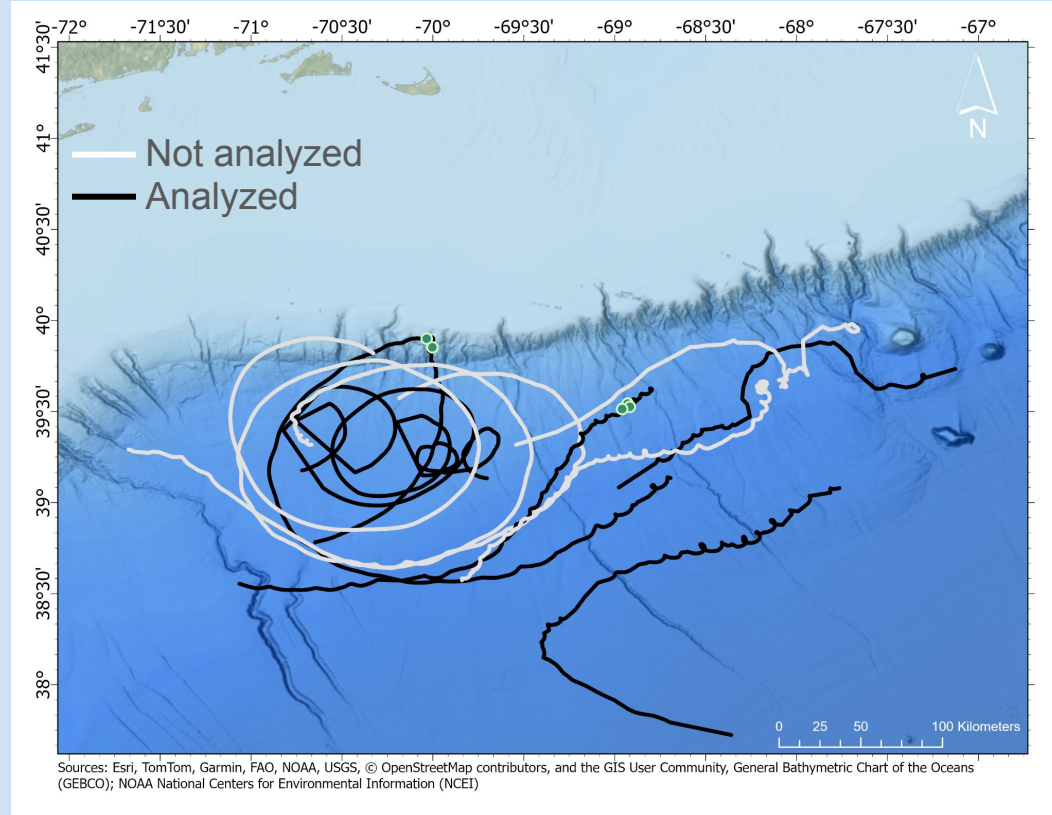
Baleen whales



6/9 drifts analyzed

Only fin whales present so far
on 2/6 drifts

At canyons and abyssal plain



Beaked whales - 202508 (late-summer)

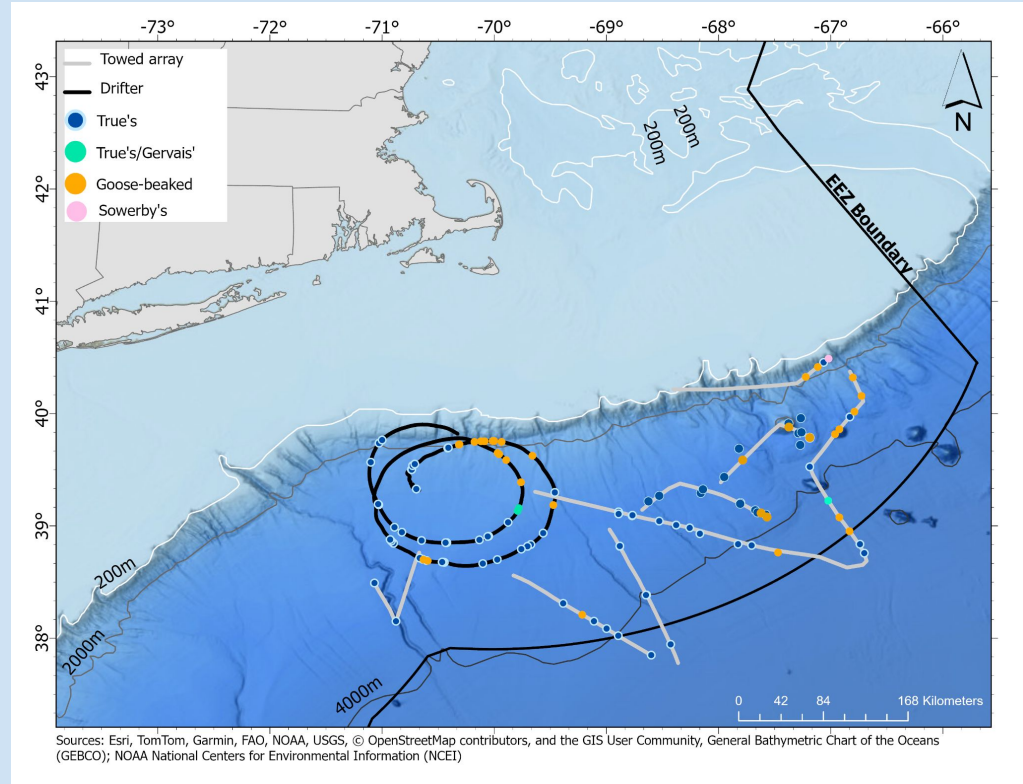


1/9 drifts complete

6 towed array deployments
complete

3 species present

Commonly found



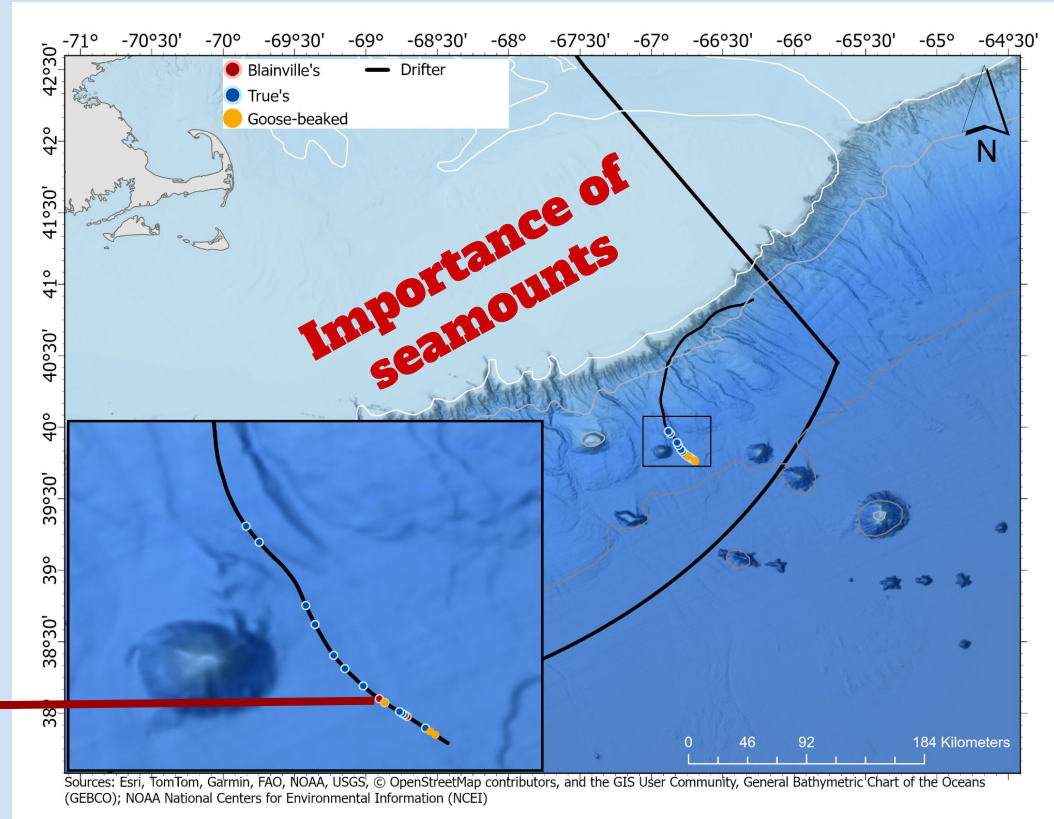
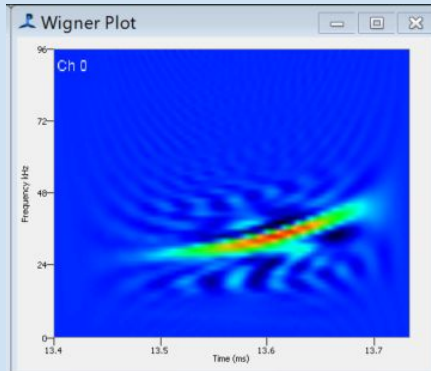
Beaked whales - 202605 (late spring)



17% of 1 drift

3 species present

Blainville's

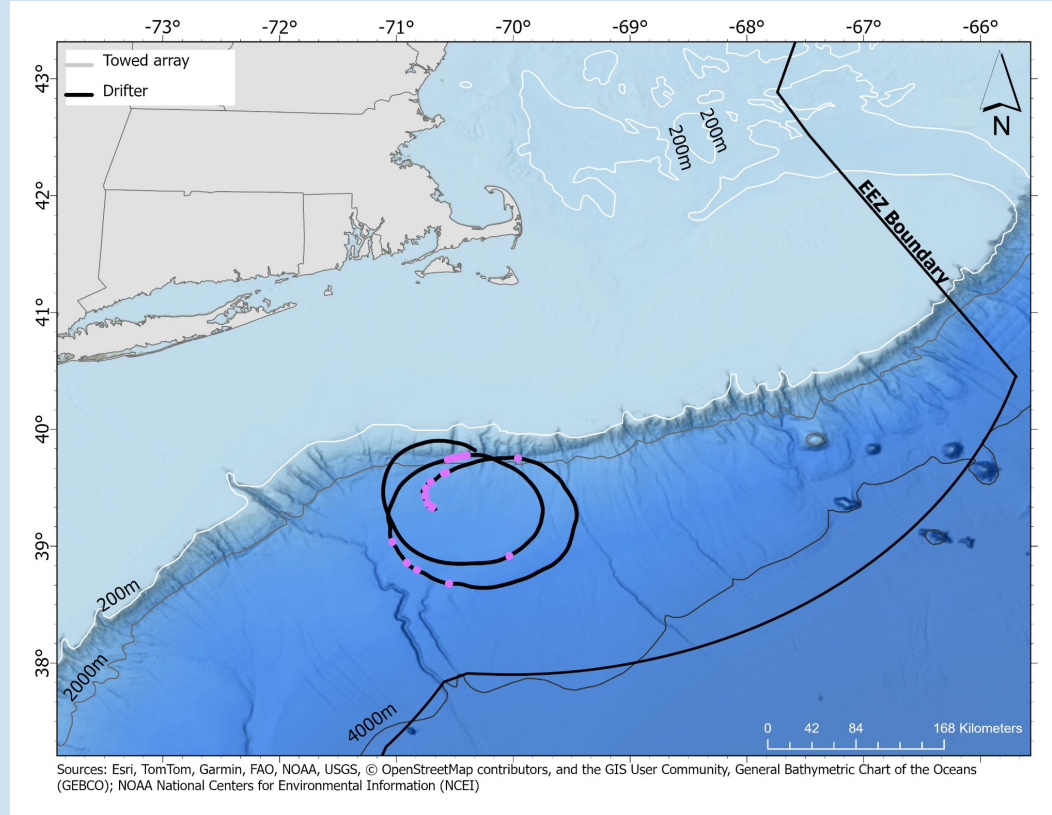


Kogia - 202508 (late summer)



1/9 drifts

Intermittently present



Future goals

Each trip building on from the last one

- More towed array transects

- Visual surveying when possible

- Tuna larvae sampling

Expand analytical products

- Telemetry data

- Build a beaked whale classifier for FPODs

- Temperature data



Adapting for other areas - things to consider



- What are the major currents in your study area?
 - Currents that show up red or white in Windy to be avoided unless staying out at sea for a few days
- Drogues do not seem to slow down drifts
- Apply the Novabraid weak rope to maintain structural stability while being whale safe
- What are other sensors can you add to your mooring?
- SmartOne GPS by Lonestar very accurate
- Data processing can take some time and lots of computer memory



Acknowledgements

Catherine Dodge

Annie Smith

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Renea Briner

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Lindsey Transue

Tony Alvernaz & crew

Hana Koilpillai

Cameron Miele

Paige Hanson

Jennifer McCullough

Kaitlyn Ho

Shannon Rankin

Jessica McCordic

Cory Hom-Weaver

Rochelle Gordon

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